DONALD HAVERY

AI Systems Builder | Full-Stack Developer | Enterprise SaaS Architect

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PROFESSIONAL SUMMARY

AI systems builder and full-stack developer with 37 GitHub repositories, 1,372+ commits, and 888K+ lines of production code across enterprise SaaS, real-time AI voice assistants, multi-agent orchestration platforms, and AI-powered business tools. Designed and shipped FuelFleet — an enterprise SaaS platform for mobile fuel delivery with 6 role-based portals, 9 access tiers, real-time fleet tracking, AI route optimization, and cloud deployment across Railway and Neon PostgreSQL. Independently architected Aether — a production AI voice assistant with 7-agent orchestration, 5-model LLM routing, real-time avatar animation, and 2,000+ automated tests. Built Citadel (AI operations platform), Cortex (autonomous media cataloger), Herald (customer operations), Sentinel (compliance automation), and Vaultwise (knowledge management) — each a production-grade FastAPI application with full test suites. Was building custom AI agents, persistent memory systems, and autonomous development workflows before these became industry features. Brings 30 years and 3.5M+ miles in transportation operations — from CDL training and regional dispatch supervision to line haul freight — the rare combination of enterprise domain expertise and hands-on AI engineering that organizations cannot find in traditional candidates.

TECHNICAL SKILLS

AI & Machine Learning Claude (Sonnet/Opus), Google Gemini (Pro/Flash), Ollama, Qwen, OpenAI Whisper, HuggingFace, PyTorch, ONNX Runtime, ChromaDB, LangGraph, FLUX.1-dev, diffusers, transformers, DreamBooth LoRA, QLoRA fine-tuning

AI Agent Architecture Multi-agent orchestration, intelligent intent routing & classification, LLM function calling & tool use, RAG (Retrieval-Augmented Generation), vector search (BM25 + dense), prompt engineering, context window management, skill/tool registries, autonomous development loops

Frontend React 18, TypeScript, JavaScript (ES6+), Vite, Recharts (data visualization), Leaflet (interactive maps), Socket.io (real-time), React Router, CSS Modules, custom design systems, responsive design

Backend Python (asyncio, FastAPI), Node.js, Express.js, Prisma ORM, REST API design, WebSocket servers, JWT authentication, bcrypt, Helmet security, rate limiting, Winston logging, CORS

Cloud & Deployment Vercel, Railway, Neon PostgreSQL, Docker Compose, GitHub Actions CI/CD, custom domain configuration, SSL/TLS, environment-based configuration

Mobile & Desktop Kotlin (Jetpack Compose, Android), PySide6 (Qt6, Windows), WebRTC, Firebase Cloud Messaging, biometric authentication (Windows Hello, Face/Fingerprint), Tailscale VPN

Voice & Audio AI Faster-Whisper STT, ElevenLabs Scribe v2 (real-time STT), Kokoro TTS, Chatterbox voice cloning, ElevenLabs TTS, Silero VAD, Picovoice Porcupine wake word, ECAPA-TDNN speaker verification, echo cancellation

Computer Vision & Avatar InsightFace/ArcFace, LivePortrait, Audio2Face-3D ONNX, Unreal Engine 5 MetaHuman, Pixel Streaming, BLIP captioning, Moondream, Qwen2.5-VL

Data & Infrastructure ChromaDB (vector DB), SQLite, PostgreSQL, Redis, Prisma ORM, Peewee ORM, Google APIs (Drive/Gmail/Calendar), Firebase, Git (1,372+ commits)

Hardware & IoT Raspberry Pi, Arduino, sensor integration, embedded systems prototyping, hardware-software interface development

PROJECTS PORTFOLIO

Aether — Production AI Voice Assistant (2 Generations)

github.com/dbhavery/aether | Python | 1,372+ commits | v1: 152K+ LOC, v2: 9.5K LOC | 2,000+ automated tests

Built a complete real-time AI voice assistant from scratch across two generations — original architecture, multi-agent orchestration, real-time voice processing, photorealistic avatar animation, and cross-platform deployment to desktop, Android, Telegram, and WebSocket.

Multi-Agent Orchestration - Designed **OrchestratorAgent + 7 sub-agent architecture** (ConversationNode, GeminiNode, ClaudeNode, ToolNode, AgentNode, VisionNode, MemoryNode) — each agent specialized for its domain, coordinated by a central orchestrator with timing, filler generation, and barge-in detection - Built **5-model intelligent LLM routing** with multi-layer intent classification: instant regex match, user override detection, keyword patterns (77 phrases), real-time context, LLM classification fallback (local Qwen, 50-100ms), and automatic tier downgrade on failure — routing across local Qwen (91-380ms), Claude Sonnet/Opus, and Gemini Pro/Flash - Implemented **32 core tools and 509 skills** with dynamic Gemini function declarations, fuzzy matching, agentic tool-calling loops (max 5 rounds with result injection), and usage analytics - Built v2 clean rewrite in **1,372+ commits over 3 days** — modular EventBus architecture with backpressure, 16 modules (12 core + 4 extensions), typed event system, and security-first design with 2 completed audit waves

Real-Time Voice Pipeline - Engineered **449ms p50 latency** (microphone to first audio output): Silero VAD, Faster-Whisper STT (82ms) with ElevenLabs Scribe v2 real-time fallback (30-80ms), intent routing, LLM streaming, sentence-by-sentence TTS synthesis via Chatterbox Turbo voice cloning — competitive with commercial voice assistants - Implemented **Picovoice Porcupine wake word** detection (92.5% accuracy), speaker verification with dynamic enrollment (ECAPA-TDNN, tiered confidence, EMA updates, 500-sample LRU), and process watchdog with auto-restart on health check failures

Memory & Intelligence - Built **tiered memory system**: 5GB LRU hot cache (v1) / 512MB (v2), gzip-compressed cold storage, ChromaDB RAG with hybrid search (BM25 + dense vector via nomic-embed-text-v2-moe), SQLite knowledge base with SHA-256 dedup, and AdaptiveLearner tracking user preferences, corrections, and barge-ins for continuous routing improvement - Created **LangGraph agent orchestration** (v2) with research and writing sub-agents, daily persona interview system, memory correction detection, and busy-state detection

Real-Time Avatar Animation - Created **Audio2Face-3D ONNX pipeline** (98 FPS on CPU) generating 52 ARKit blendshapes via diffusion model, streamed through custom TCP binary protocol to UE5 MetaHuman with Pixel Streaming video delivery - Built **Ditto motion pipeline**: HuBERT audio encoding (1024-dim @ 25fps) feeding LMDM conditional diffusion transformer for 265-dim motion synthesis, rendered through LivePortrait warping network — 20-23 FPS at 3.3GB VRAM, 98-100% lip sync detection - Implemented **LivePortrait integration** with precomputed keypoint ring buffer achieving 23+ FPS at 512x512 on RTX 3090 Ti, MJPEG stream relay to desktop client

Cross-Platform Deployment - **Windows Desktop**: PySide6 neumorphic UI with 3 views (text chat, voice call, full-screen video), Windows Hello biometric auth, settings panel with YAML round-trip config sync, system tray integration, always-on-top mode - **Android**: Kotlin/Jetpack Compose with WebRTC video, biometric auth, FCM push notifications, Tailscale network-aware auto-switching (WiFi/cellular), real-time conversation sync - **Telegram Bot** and **WebSocket Server** with binary protocol (14 message types), token authentication, rate limiting, 10-client connection limit

Security & Quality - Built **three-layer security**: command injection blocking (11 regex bypass patterns), PurePosixPath path traversal protection, permission framework, WebSocket token auth (32-byte hex, constant-time verify), PC control allowlists, prompt injection defense ([EXTERNAL_CONTENT] trust boundaries) - Managed **GPU coordination** across 7+ models within 24GB VRAM: priority-based loading, lazy initialization, tensor offloading, automatic idle unloading - Wrote **2,000+ automated tests**: unit, integration, performance benchmarking, chaos/failure injection, end-to-end — with pytest, Ruff linting, Pyright type checking, Bandit security scanning, and GitHub Actions CI/CD

Citadel — AI Operations Platform

github.com/dbhavery/citadel | Python | 6 packages | 10K+ LOC | 100% test coverage

- Built modular monorepo for production AI infrastructure with **zero-dependency HNSW vector search** — eliminates $70+/mo managed vector DB costs with pure Python implementation
- Designed **multi-provider LLM routing** with automatic failover across Claude, Gemini, and Ollama with semantic caching that saves $0.01-0.10 per cached hit
- Implemented **ReAct agent framework** with tool execution, thought chains, and multi-step reasoning across 6 independently deployable packages

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FuelFleet — Enterprise SaaS Platform for Mobile Fuel Delivery

github.com/dbhavery/fuelfleet-platform | 12 repositories | 1,372+ commits | 19K+ LOC | 19K+ lines CSS React 18, TypeScript, Node.js, Express, Prisma, PostgreSQL, Socket.io, Leaflet, Recharts | Deployed on Railway + Vercel

Designed, architected, and built a complete enterprise SaaS platform for a mobile fuel delivery business from concept to cloud deployment — 6 portals, 9 user roles, real-time fleet tracking, AI route optimization, and a 255-input simulation engine. This is not a tutorial project; it is a production-ready startup platform with custom domains, role-based access control, and cloud infrastructure.

Platform Architecture - Architected **microservices ecosystem** with 12 repositories: 6 independent portals, shared design system, core API engine, simulation engine, and deployment infrastructure — each service independently deployable with shared authentication and design tokens - Built **Core Engine API** (Express.js + Prisma ORM + Neon PostgreSQL) with JWT authentication, bcrypt password hashing, Helmet security headers, rate limiting, file uploads (multer), structured logging (Winston), CORS management, and real-time event streaming (Socket.io) - Deployed **full cloud infrastructure** on Railway with custom domain routing, auto-generated SSL certificates, and PostgreSQL via Neon connection pooling

6 Role-Based Portals - **HQ Command Center** (Holdings Admin) — 18-page admin dashboard with financial analytics, franchise monitoring, marketing campaign management, legal/compliance tracking, SBA lending metrics, live website content editor, and real-time fleet maps (React-Leaflet) with KPI dashboards (Recharts). Secondary authentication gates on sensitive areas - **Franchise Portal** — Dual-mode interface: P&L Cockpit for franchise owners and Mission Control for managers (daily operations, delivery scheduling, driver/vehicle management, fuel inventory) - **Driver Co-Pilot** — 4-mode operational interface (Parked, Driving, Delivery, Resupply) with real-time route optimization, customer eligibility checks, fuel tank monitoring, safety alerts - **Customer Portal** — 5 sub-portals: Residential, B2B Employer, B2C Employee, Investor, SBA Lender - **Unified Auth Gateway** — Centralized authentication across all portals

Simulation & Testing Engine - Engineered **255-input Simulation Engine** (TypeScript) generating realistic data across 20 categories — vehicle GPS/telemetry, fuel dispensing, AI safety camera feeds, driver performance, payment processing, environmental conditions, equipment maintenance, and anomaly detection - Built 9 scenario presets with time control (pause/resume/1x-100x speed), per-vehicle WebSocket subscriptions, manual sensor override injection

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Cortex — AI Media Cataloger

github.com/dbhavery/cortex | Python (FastAPI) | 164 tests | 28 file types

- Built autonomous media indexing system with **BLIP captioning**, **InsightFace face recognition** (512-dim ArcFace embeddings with DBSCAN clustering), and Google Photos/Drive sync — processes 5,700+ files with zero manual intervention
- Designed continuous processing pipeline — drop a file in, it's captioned, face-detected, categorized, and searchable within 60 seconds
- Content-hash deduplication (SHA-256) handles renames, copies, and cross-source duplicates across local and cloud storage

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Herald — Customer Operations Platform

github.com/dbhavery/herald | Python (FastAPI) | 95 tests | 34 endpoints

- Built AI-powered customer ops platform reducing support response time from minutes to seconds with **citation-backed AI responses** and pluggable LLM backend
- Implemented **TF-IDF weighted lexicon sentiment scoring** with IDF corpus weighting, 6-category auto-categorization, and 5-factor customer health scoring (activity, sentiment, tickets, plan, tenure)
- Designed churn prediction and SLA analytics with real-time dashboards

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Sentinel — Compliance Autopilot

github.com/dbhavery/sentinel | Python (FastAPI) | 50 tests | 62 requirements | 14 templates

- Built automated compliance platform for **SOC 2, GDPR, HIPAA** — gap analysis in seconds vs weeks of manual audit
- Implemented AI policy generation, document scanning, drift monitoring, and audit-ready report generation with configurable framework templates

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Vaultwise — Knowledge Management Platform

github.com/dbhavery/vaultwise | Python (FastAPI) | 133 tests | 20 endpoints

- Built **TF-IDF search engine from scratch** — zero ML dependencies, pure Python tokenization, stop word removal, TF/IDF calculation, L2 normalization, and cosine similarity
- Designed document ingestion with word-based chunking, AI-powered Q&A with source citations, and auto-generated training articles and quizzes

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Custom Image Generator & Editor (CIGE)

github.com/dbhavery/Custom-Image-Generator-and-Editor | Python/PyTorch | 127 tests

- Built local AI image generation — **$0 per image** vs $0.02-0.10 on cloud APIs. FLUX.1-dev for quality, RealVisXL Lightning for speed
- Implemented VRAM-aware model management with neumorphic desktop UI (PySide6), 3 generation modes, and CUDA-optimized inference on RTX 3090 Ti

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Autoloop — Autonomous Development Loop

github.com/dbhavery/autoloop | Bash | 573 BATS tests

- Built hands-free task execution wrapper for Claude Code with **dual-condition exit gate** — requires both heuristic completion indicators AND explicit exit signal to prevent premature stops
- Implemented **circuit breaker pattern** detecting stagnation (3 loops no changes), repeated errors (5 loops same error), and output decline (>70% drop) with automatic recovery
- Three-layer API limit detection (timeout guard, structural JSON, filtered text fallback) — zero dependencies beyond Bash 4.0+ and jq

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AI Developer Tools

github.com/dbhavery | Python | 5 tools | 196 tests

- **fineforge** — Fine-tune LLMs on consumer GPUs at $0 cloud cost (48 tests)
- **promptlab** — Test prompts across 3+ LLM providers with assertion framework (55 tests)
- **ragtest** — RAG pipeline evaluation and benchmarking (54 tests)
- **mcplex** — MCP server for local AI models (24 tests)
- **coderev** — AI-powered code review GitHub App using Claude (15 tests)

Additional Projects

Aether Avatar (github.com/dbhavery/aether-avatar) — 8-module offline pipeline for photorealistic avatar generation. Face identity: DreamBooth LoRA on FLUX.1-dev with 90-image dataset, 0.71-0.79 cosine similarity.

OpenClaw — Production Telegram AI agent with QMD memory backend (BM25 + vector semantic search + LLM re-ranking), local Qwen 8B inference.

Hardware & IoT — Raspberry Pi and Arduino projects with sensor integration, custom automation — early foundation for systems-level thinking applied to AI development.

PROFESSIONAL EXPERIENCE

Old Dominion Freight Lines — Portland, OR

Transportation Operations Specialist (Line Driver) | 2022 – Present - Execute line haul freight operations across the Pacific Northwest for one of the nation's largest LTL carriers, managing load integrity, DOT compliance, and on-time delivery across multi-stop routes - Coordinate with dispatch, dock operations, and terminal teams in real-time — the same distributed systems coordination applied to multi-agent AI orchestration and WebSocket event pipelines - Maintain CDL Class A with Hazmat, Tanker, and Doubles/Triples endorsements, operating combination vehicles under strict FMCSA safety regulations - Apply systematic problem-solving to route disruptions, equipment issues, and weather conditions — complex, time-critical decision-making with incomplete information

Peninsula Freight Lines — Portland, OR

Transportation Operations Specialist (Line Driver) / **Driver Trainer** | 2017 – 2021 - Performed line haul and regional freight operations, maintaining consistent safety and on-time performance records - Selected as certified driver trainer — assessed skills, provided road instruction, and ensured safety compliance for new drivers - Served as bridge between drivers, dispatch, and terminal operations — translating operational complexity into actionable execution

Alpha Transport — Portland, OR

Operator | 2012 – 2017 - Operated commercial vehicles across Western states routes, building the foundational logistics experience that informs current AI systems architecture - Maintained safety compliance and equipment standards across long-haul operations

Earlier Career — Various Companies

Supervisor, Dispatcher, Driver Trainer, Operator | ~1996 – 2012 - Held supervisory positions managing teams of 5–6 across daily freight operations - Dispatched regional areas coordinating dozens of employees, routes, and schedules - Trained dozens of drivers to successfully obtain their CDL certifications - Accumulated 3.5M+ career miles of safe commercial driving across 30 years in the transportation industry

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CERTIFICATIONS & LICENSES

Transportation	AI & Technology
CDL — Class A	37 GitHub repositories, 1,372+ commits, 888K+ LOC
Hazmat Endorsement (H)	Enterprise SaaS deployed to Railway + Vercel
Tanker Endorsement (N)	2,000+ automated tests across projects
Doubles/Triples Endorsement (T)	Multi-agent AI system architecture
DOT Compliance & Safety Certification	Full-stack web development (React + Node.js + PostgreSQL)
OSHA Safety Training	Real-time systems & WebSocket engineering
Smith System Defensive Driving Instructor	CI/CD pipeline design (GitHub Actions)
Certified Transportation Manager	GPU resource management & ML inference optimization
Dispatch Certification	Cloud deployment & infrastructure (Railway, Vercel, Neon)
Freight Broker License/Authority	1,000+ hours self-directed AI/ML education
ELD & Fleet Management Systems	**DeepLearning.AI — Generative AI with LLMs** (Coursera, 2026)
Certified Driver Trainer	*Currently pursuing:* Google Cloud Professional ML Engineer, AWS ML Engineer Associate, Databricks GenAI Engineer
Logistics & Supply Chain Certification	

EDUCATION

High School Diploma Capital High School — Boise, ID | 1996

Continuous Professional Development - 1,000+ hours self-directed AI/ML curriculum: LLM architecture, multi-agent systems, prompt engineering, Python, JavaScript/TypeScript, full-stack development, mobile development, computer vision, speech processing, model fine-tuning, cloud deployment, hardware integration - Build-first methodology: every concept studied was immediately applied in production code — 37 repositories and 1,372+ commits of working, deployed software - Hardware foundations: Raspberry Pi and Arduino sensor/automation projects before transitioning to AI software development